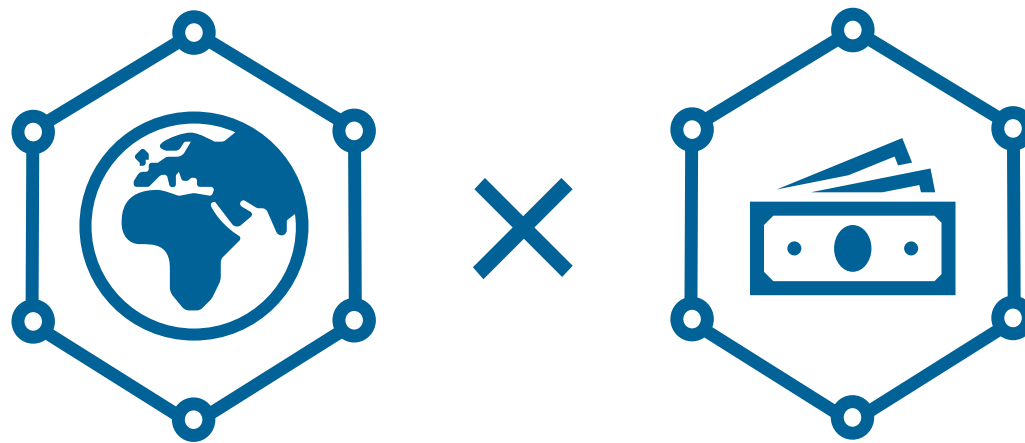
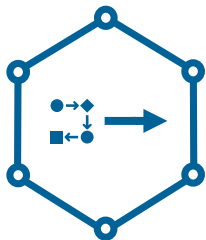
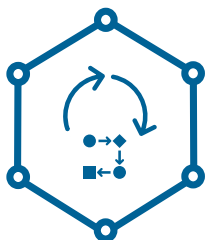
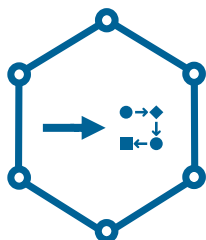




Financial Knowledge Graphs

Company Knowledge Graphs

Emanuel Sallinger



Ongoing work on the reference historical KG for all Italian companies, including:



- 6 million **companies**
- 10 million **people**
- 30 million **ownerships**
- 20 million **roles** (e.g. managers)
- 200k company **events** (like M&A)



- 50+ million **nodes**
- 100+ million **edges**
- 1+ billion **properties**

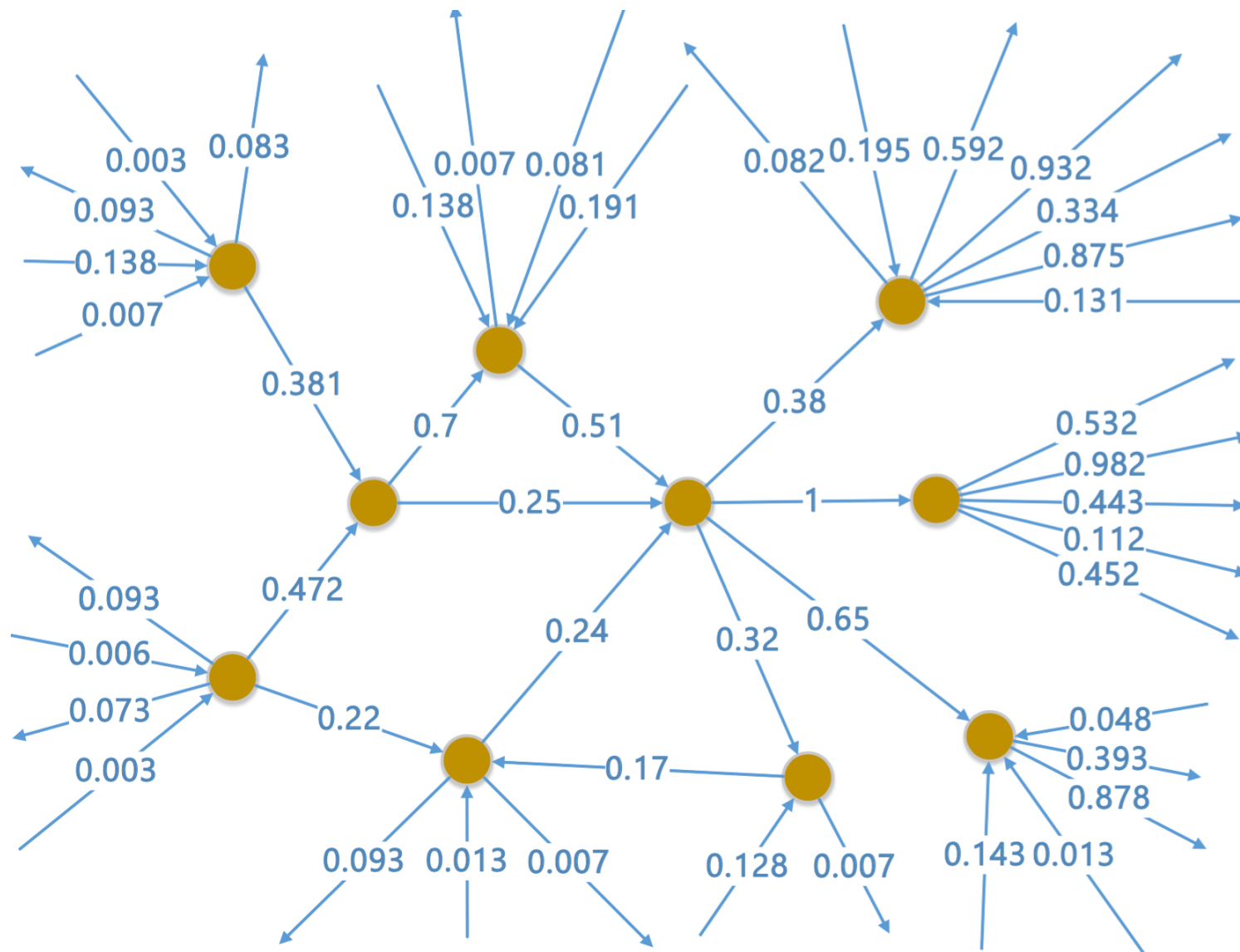
plus, adding **derived** extensional knowledge:



- company **control**
- **family links** between people



Ownership Graph





Creditworthiness

- Probability that a borrower company will not default on its debt obligations

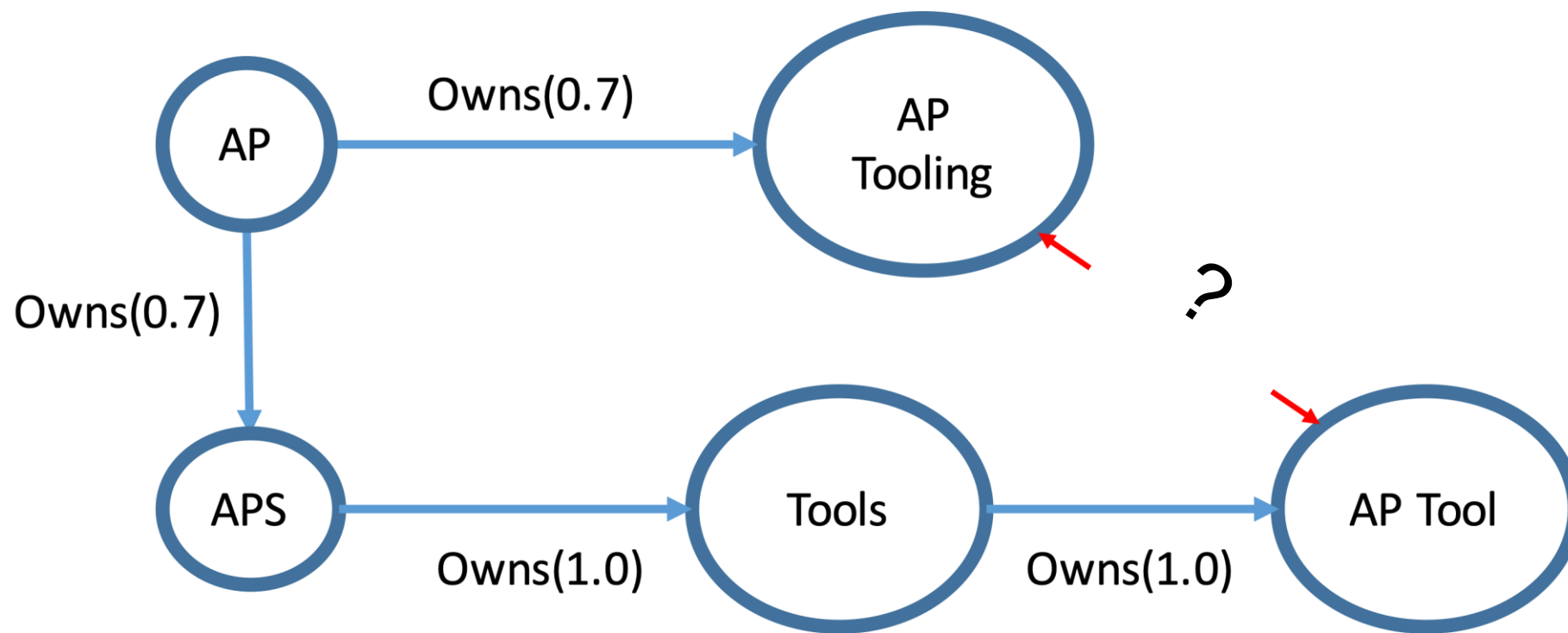
Credit DB				
Company name	Total Debts	...	Avg Debt to Income Ratio	Rating
AP Tool	0	...	0.28	A
ACME	50k	...	0.46	B
Big S	20k	...	0.89	D
AP Tooling	0	...	0.10	A

LOAN DENIED

AP Tool asks for 1M loan, with AP Tooling as guarantor

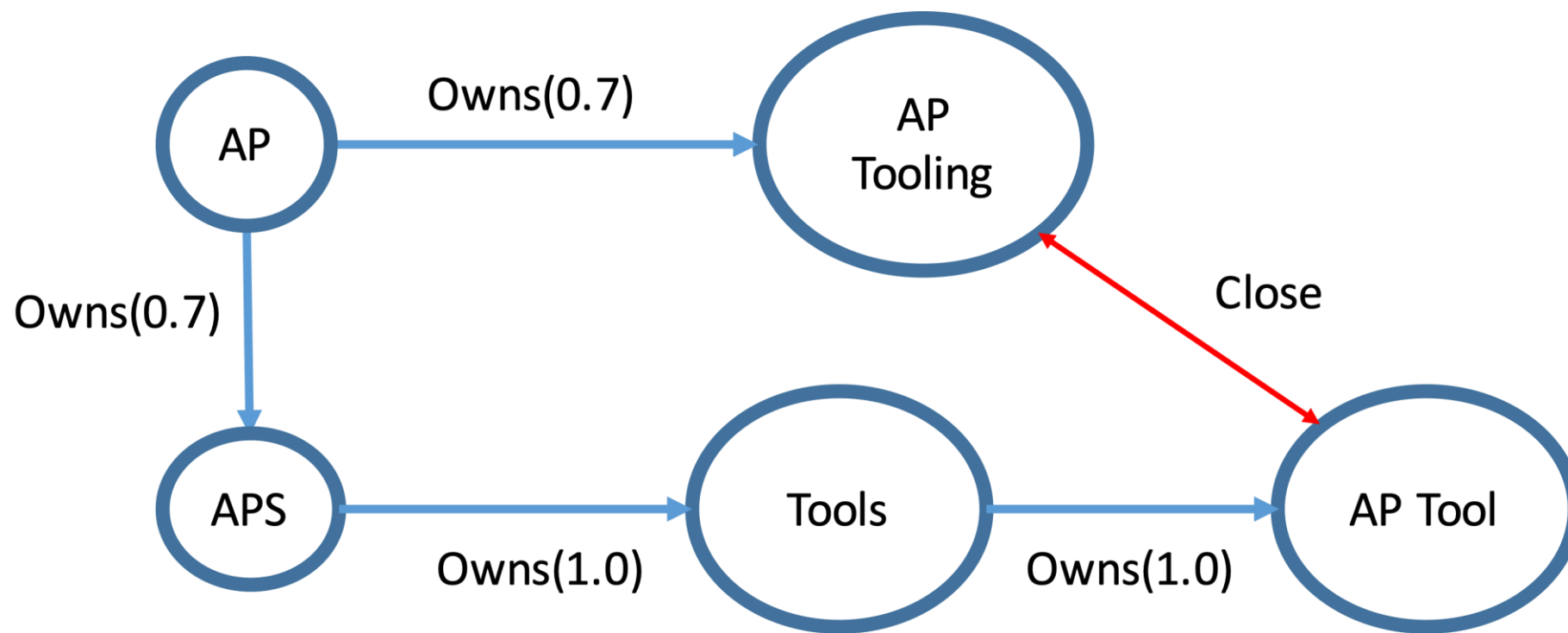


Company Graphs





Company Graphs

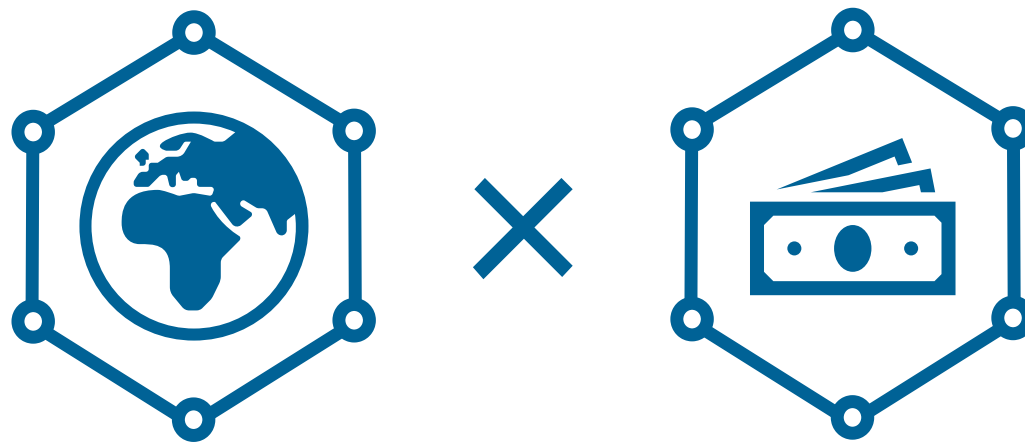




Company Graphs

We build graphs of company networks, to:

1. reveal **power**
 1. finding **controllers**
 2. studying the **structure** of market
 3. studying **dispersion** of control
 4. global **shareholding** analysis
 5. intercept and prevent **hostile takeovers**
2. detect **collusion** and do **forensics**
 1. support **AML**
 2. detecting **ultimate beneficial owners**
3. evaluate and guarantee **anonymity** of **microdata**
4. evaluate **risks**
5. model **propagations** (e.g., of shocks)
6. guarantee **compliance**
7. perform enhanced **due diligence**
8. understand **foreign shareholder structures**
9. know real **cash flows**



Financial Knowledge Graphs

Company Knowledge Graphs

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